R13

Code No: 113AU

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, November/December - 2018
ELECTRONIC DEVICES AND CIRCUITS
(Common to EEE, ECE, CSE, EIE, IT, MCT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

JNTUH USED 28-11-2018 AM

PART- A

(25 Marks)

1.a) Draw the V-I characteristics of varactor diode. [2]
b) Explain how does the reverse saturation current of a p-n diode vary with temperature. [3]
c) Define the percentage of regulation. [2]
d) What is the need of rectifier? List different types of rectifiers. [3]
e) Illustrate CE cutoff region. [2]
f) Explain the base spreading resistance. [3]
g) What is thermal runaway? [2]
h) List and briefly explain three sources of instability of collector current. [3]
i) What are the different types of FETs? [2]
j) What is pinch-off voltage? Give its expression. [3]

PART-B

(50 Marks)

2.a) Derive the expression for transition capacitance C_T of a diode.
b) Draw the small signal model of Tunnel Diode operating in the negative resistance region. [5+5]

OR

3.a) Draw and explain the graph indicating the variation of minority carrier density with distance in a p-n junction diode under forward biased condition.
b) Write the equation for Volt-Ampere characteristics of a photo diode. [5+5]

4.a) Draw and explain the circuit of a half-wave rectifier with capacitor filter.
b) Explain the design of Full wave bridge rectifier. [5+5]

OR

5.a) A half wave rectifier has a load of 3.5 KΩ. If the diode resistance and the secondary coil resistance together have a resistance of 800Ω and the input voltage has a signal voltage of 240 V, calculate i) Peak, average and rms value of current flowing. ii) dc power output. iii) ac power input iv) Efficiency of the rectifier.
b) Compare inductor and capacitor filters. [6+4]

6.a) Explain the input and output characteristics of a transistor in CB configuration.

b) Draw the circuit diagram of an NPN junction transistor in CE configuration and describe its characteristics. [5+5]

OR

7.a) Illustrate and explain active region, saturation region and cut-off region in transistor characteristics.

b) Explain the hybrid small signal model for common collector configuration. [5+5]

8.a) Draw a transistor amplifier using self bias (CE configuration) and explain the operation.

b) Discuss the stabilization against variations in V_{BE} and β. [6+4]

OR

9.a) Draw and explain the Collector feedback biasing.

b) Describe the analysis of Transistor Amplifier circuit using h-parameters. [6+4]

10.a) Explain how FET works as voltage variable resistor.

b) Compare BJT and FET. [6+4]

OR

11. Explain the constructional features of a depletion mode P-channel and Enhancement mode MOSFET and explain its basic operation. [10]

---ooOoo---

JNTUH UCEV 28-11-2018AM